Workplace parking pricing and commuting mode

Student project guidelines

1 Context

A Swiss public employer operates two sites: a main site in **Lausanne (VD)** and a secondary site in **Sion (VS)**. Both sites charged staff CHF 2 per day for on-site parking in 2024. In 2025, the Lausanne fee increased to CHF 4, while the Sion fee remained CHF 2. No other mobility policy changed at either site.

All 1,000 employees were surveyed in both years. The employer must now choose its 2026 parking tariff. Your group will act as a policy-analysis team and advise both the mobility manager and the staff council.

Your analysis will be guided by three linked questions:

1. How many people stop driving when the price doubles?
2. What would happen at CHF 6 or CHF 8?
3. Who ends up paying?

The objective is to build a transparent empirical argument from raw data to a defensible policy recommendation.

2 Data file

`parking_data.csv` contains 2,000 rows: 1,000 employees observed in 2024 and 2025.

Variable	Type	Description
`employee_id`	string	Anonymised identifier, stable across years
`site`	string	`Lausanne` or `Sion`
`year`	int	2024 or 2025
`parking_fee_chf`	float	Daily fee in force at that site and in that year
`main_mode`	string	`car`, `bus`, or `bike`
`uses_car`	int	1 if `main_mode == "car"`; 0 otherwise
`car_days_per_week`	float	Average days driven to work, from 0 to 5; 0 for non-car commuters
`home_commune`	string	Municipality of residence
`home_postcode`	int	Swiss postcode
`distance_km`	float	Road distance from home to the work site
`age`	int	Age
`gender`	string	`F` or `M`
`n_children`	int	Children under 16 in the household: 0, 1, or 2, where 2 means "2 or more"
`fte`	float	Full-time equivalent contractual workload: 0.6, 0.8, or 1.0
`gross_income_chf`	int	Annual gross income at the employee's observed workload

You can assume there are 46 working weeks in a year.

3 Basic Empirical framework description

3.1 Two equations

The project separates two margins of car use:

(1) the **extensive margin**: whether the employee's main commuting mode is car (mode choice); and

(2) the **intensive margin**: how many days per week a car user drives (frequency).

3.2 Equation (1): mode choice

A general linear probability model is:

$$\texttt{uses_car} = a \cdot \log(\texttt{fee}) + \text{controls} + \log(\texttt{fee}) \times \text{interactions}, \tag{1}$$

Equation (1) is a linear probability model. Its policy effects are reported in probability units or percentage points. The interaction terms should comprise the transit index (Part A), income and number of children.

You may represent the policy through log fee or through the Lausanne–2025 interaction in a site-and-year comparison model (i.e. $(L_i \times \text{Post}_t)$, where L identifies lausanne and Post identifies the year). With the same sample, controls, and interactions, these are equivalent parameterizations of the same policy variation, but the coefficients have different units.

3.3 Equation (2): frequency

Among car users only, estimate:

$$\log(\texttt{car_days_week}) = b \cdot \log(\texttt{fee}) + \text{controls} + \log(\texttt{fee}) \times \text{interactions}. \tag{2}$$

A coefficient on log fee is directly an elasticity because both the positive outcome and the fee are logged. The interaction terms should comprise the transit index (Part A), income and number of children.

Part A — Transit accessibility

For each `home_commune`, construct a measure of how realistic public transport is for commuting to the relevant site. Use published timetable information from appropriate operators. At a minimum, collect:

- public-transport travel time relative to car travel time for arrival around 08:00;
- the number of departures in the 06:00–09:00 window; and
- the time of the first feasible arrival at the work site.

Combine these dimensions into a continuous score or a small number of ordered categories. Explain the coding rule and procedure.

Required output:

- **Table A1:** one row per commune, or a compact summary if there are many communes, showing the underlying transit indicators and final classification.
- A machine-readable file named `transit_access.csv`, containing the complete commune-level measure, source links or source identifiers, and coding notes if necessary.
- A short validation paragraph explaining whether the resulting ranking is plausible and identifying at least one limitation.

Part B — Extensive margin: mode choice

1. Cross-tabulate `main_mode` by site and year. Report the **gross flows in both directions** (the amount of people switching from car to another mode, and the opposite) and the net mode share percentage change.

   Required output: Table B1 with mode shares by site and year and the numbers switching car-to-non-car and non-car-to-car.

2. Compute the change in the car share in both **percentage points** and as an **elasticity**. Explain why these answer different questions, and identify which measure belongs in the policy memo.

3. Describe heterogeneity in these results: disaggregate the previous result by

   - transit-accessibility group
   - income tercile
   - number of children

   Define how groups are formed and report group sample sizes.

   Required output: Table B2 with, for every requested group, 2024 and 2025 car shares, the percentage-point change, extensive elasticity, and number of employees.

4. Estimate equation (1) sequentially:

   (a) a Lausanne-only before–after or price specification;

   (b) a specification adding Sion as a comparison site;

   (c) the comparison specification with a justified set of controls, including transit accessibility; and

   (d) a specification allowing the policy response to differ with transit accessibility, income, and number of children (interactions).

   State how the variables entering the interactions are coded or grouped. You may, for example, mean-centre a continuous variable by subtracting its sample mean; in that case, the main policy coefficient represents the estimated response for an employee with average values of the centred characteristics. If you use categories instead, state the reference group. Interpret the interaction pattern and compare it with Table B2; a separate table of predicted group effects is optional.

   Required output: Table B3, a single four-column regression table. Report policy variables, interaction terms, and other controls.

   Limit the Sion comparison to **one short paragraph**. State what it contributes, how it changes the Lausanne-only result, and one reason for caution. Do not turn the report into an extended discussion of difference-in-differences.

Part C — Intensive margin and decomposition

The intensive margin is computed only among car users, for whom `car_days_per_week`>0. Total margin is among all employees.

1. Compute the overall intensive elasticity from the change in mean driving days among car users. Then describe how this response differs by transit-accessibility group, income tercile, and number of children. State whether these are descriptive subgroup estimates or model-implied effects.

Required output: Table C1 with 2024 and 2025 mean car days among car users, intensive elasticity, and sample size for the full sample and each requested group.

2. Estimate equation (2) sequentially: Lausanne only; then with Sion; then with controls; and finally with controls and interactions. Use the same substantive logic as in Part B.

   Required output: Table C2, a compact regression table with the four specifications. Report the policy coefficient in its native units and, where necessary, the corresponding elasticity. Include uncertainty, observations, R^2, and sample restrictions.

3. Compare the estimates across specifications. Discuss selection into the car-user sample: is the average car user in 2025 the same kind of person as the average car user in 2024? Use descriptive evidence or covariate comparisons.

4. Verify the decomposition using mutually consistent definitions and samples:

$$\varepsilon_{\text{total}} = \varepsilon_{\text{extensive}} + \varepsilon_{\text{intensive}}.$$

   State which margin accounts for most of the total change.

   Required output: Table C3 reporting the extensive, intensive, and total elasticities, together with the underlying means used in the calculation.

Part D — 2026 tariff scenarios

Forecast outcomes at daily fees of CHF 5, CHF 6, CHF 8, and CHF 10. Choose the elasticity estimates used for the central forecast and justify that choice. Use a coherent mapping from elasticities to outcomes, and do not extrapolate silently.

At minimum, report predicted:

- car share;
- days driven per car user;
- total car days per employee; and
- annual parking revenue.

Required outputs:

- **Table D1:** one row per tariff, including the 2025 CHF 4 baseline, with all four outcomes above.
- **Figure D1:** one clearly labelled scenario figure showing the main behavioural outcome and revenue across tariffs.
- A sensitivity check using at least one defensible alternative elasticity or specification.
- A short paragraph stating the assumptions, the credible range of extrapolation, and the main reason forecasts may fail outside that range.

Part E — Distribution and incidence

For each transit-accessibility group, income tercile, and children group, calculate annual parking expenditure as a share of observed gross income:

(a) averaged over all employees in the group, including non-drivers; and

(b) averaged over employees who still drive in 2025.

The first quantity describes average exposure across the group; the second describes the burden among those who actually continue paying. Report the two quantities separately. Where feasible, also show the projected burden under your recommended 2026 tariff, making clear which behavioural assumptions are used.

Required output: Table E1 with group sample sizes, the two 2025 burden measures, and the corresponding measure under the recommended tariff if it is projected.

Interpret the results using both responsiveness and burden. Address the following questions:

- Does a weak response indicate low burden, or could it indicate a lack of alternatives?
- Which groups obtain the largest behavioural benefit, and which groups contribute the most revenue?
- Are there groups for whom a uniform tariff creates a foreseeable equity or feasibility problem?

Do not infer welfare, savings, or public-transport costs that are not observed in the data.

Part F — Policy recommendation

Recommend a specific 2026 tariff and, where justified, complementary measures. You can be creative in the proposed solutions. The recommendation must be addressed jointly to the mobility manager and the staff council. It must balance:

- behavioral effectiveness and expected reduction in car use;
- revenue and the proposed use of that revenue;
- distributional burden and access to realistic alternatives;
- administrative and political feasibility;
- uncertainty, unintended effects, and limits of the evidence.

A strong recommendation does more than select the tariff producing the highest predicted revenue. It explains the objective being prioritized, the trade-offs accepted, and why the chosen package is preferable to at least one plausible alternative. Consider whether complementary measures, exemptions, phased implementation, or revenue recycling improve the policy; if you propose one, explain how it would be targeted and administered.

End the memo with a short implementation and monitoring plan. Identify two or three indicators to collect after implementation, when they should be reviewed, and what evidence would trigger a policy adjustment.

Required output: a three-page policy memo containing a clear decision, the central evidence supporting it (including a summary of the empirical results), the principal distributional implication, one alternative considered, and an implementation and monitoring plan.

Submission

The submission is due on the 08th of December, 23:59.

Submit the following files as one group package:

1. **Policy memo, PDF: maximum 3 pages.** Use no more than two compact tables or figures in the memo. This is intended as being an answer to the Swiss employer who contacted you to get insights.

2. **Technical report, PDF: maximum 8 pages.** The limit includes required outputs: regression tables, figures, notes, and references. Do not include a cover page or repeat the project instructions.
3. **Reproducible code:** one clearly structured Python(.py) or R(.Rmd) file. The code must recreate all reported numerical results and figures from the supplied data plus `transit_access.csv`.
4. **Transit-access file:** `transit_access.csv`, including source information and coding notes explaining the method used to compute the transit index.

Minimum technical report content

The technical report should contain Tables A1, B1–B3, C1–C3, D1, E1, and Figure D1. Tables may be combined into labelled panels when this improves readability, but required quantities must remain easy to locate. Number tables consistently and use the same variable names, samples, and rounding conventions throughout.

Do not paste raw `statsmodels`, R, Stata, or notebook output. Produce your own tables with readable labels. Regression tables must state the dependent variable, coefficient units, reference categories, standard-error or confidence-interval method, sample size, and relevant sample restrictions.

Data decisions and reproducibility

The data contain deliberate complications. Your group must identify and handle them transparently!

Any use of generative AI, code assistants, or automated writing tools must comply with course rules and be disclosed briefly in the report. The group is responsible for verifying every result, source, citation, and interpretation.